unstable isotopes. When the chemical properties of the two substances are different, and the substance is present in ponderable quantities, the two can always be separated by chemical means without the slightest difficulty. But the situation is different when dealing with imponderable quantities, as in the cases under consideration here. In these cases, adsorption phenomena frequently occur and prevent the application of the customary chemical processes of separation. The problem can often be solved by adding in advance a considerable amount of a non-radioactive isotope of the substance to be produced, to the original substance. The adsorption phenomena will then play just a negligible part, and the radioactive substance precipitates together with the stable isotopes.

The chemical processes which can be applied when handling short-lived radioactive isotopes, have been developed and perfected to a high degree by O. Hahn and his associates, in particular.

One of the most important artificial radioactive substances is radioactive phosphorus. For instance, carbon bisulphide (CS_2) is bombarded by neutrons. The following reaction will occur in the sulphur atoms:

$$_{16}S^{32} + _{0}n^{1} \rightarrow _{15}P^{32} + _{1}H^{1}$$

The sulphur atom of mass number 32, which constitutes approximately 32 per cent. of ordinary sulphur, and the neutron produce a radioactive phosphorus atom of the same mass number, plus a proton. The half-life of this radioactive phosphorus is relatively long, 14·5 days, which is obviously an advantageous feature as regards its practical utilization; it emits an electron and changes back into the original sulphur atom—viz.:

$$_{15}P^{32} \rightarrow _{16}S^{32} + _{-1}e^{0}$$

According to a suggestion by Erbacher, radioactive phosphorus is obtained simply by diluting the irradiated carbon bisulphide with water; the radioactive phosphorus is dissolved, in the form of ions, in the water which is then separated from the carbon bisulphide by one method or another.

The situation is more difficult where the radioactive substance is